

Compile-Test Result

CGI Built on=Nov 20 2023

Session=17278175911726089765

Received Files:

- yssimplesound.h
- ps5-1.cpp
- yssimplesound.cpp

Features Required:

1. OpenGL animation using double buffering:
 - a. This is achieved using the **GL colour and depth buffer**, which are periodically cleared and reset to ensure lack of lags
2. Runs the animation till the user terminates
 - a. A condition has been provided to immediately terminate the program if the user presses the ESC key
3. Must be an interactive program
 - a. The game portion of the program requires user input, and is interactive
4. Uses colour gradation for making the background in the first and last frame, to make the sky appear more realistic to a night sky
5. Uses GL_TRIANGLES, GL_TRIANGLE_FAN, etc to render the various shapes used
6. Uses euler's method to integrate the position of the cannonball during shooting. Also uses a state transition structure to navigate between different frames of the game.

In order to use the program, please compile the program with the provided yssimplesound.cpp and yssimplesound.h files (included in zip) and enjoy the game!

Notes used for audio: (from an open source coding example)

https://www.projectrhea.org/rhea/index.php/MatLab_program_that_plays_Star_Wars_theme_song